

Library_Mnagement_System

Topic : Library Management System

You are going to build a project based on Library Management System. It keeps track of all information about books in the library, their cost, status and total number of books available in the library.

Create a database named library and following TABLES in the database:

1. Branch
2. Employee
3. Books
4. Customer
5. IssueStatus
6. ReturnStatus

Attributes for the tables:

1. Branch

- Branch_no
- Set as PRIMARY KEY
- Manager_Id
- Branch_address
- Contact_no

2. Employee

- Emp_Id – Set as PRIMARY KEY
- Emp_name
- Position
- Salary
- Branch_no
- Set as FOREIGN KEY and it refer Branch_no in Branch table

3. Books

- ISBN
- Set as PRIMARY KEY
- Book_title
- Category
- Rental_Price
- Status [Give yes if book available and no if book not available]
- Author
- Publisher

4. Customer

- Customer_Id
- Set as PRIMARY KEY
- Customer_name
- Customer_address
- Reg_date

5. IssueStatus

- Issue_Id
- Set as PRIMARY KEY
- Issued_cust – Set as FOREIGN KEY and it refer customer_id in CUSTOMER table Issued_book_name
- Issue_date
- Isbn_book – Set as FOREIGN KEY and it should refer isbn in BOOKS table

6. ReturnStatus

- Return_Id
- Set as PRIMARY KEY
- Return_cust
- Return_book_name
- Return_date
- Isbn_book2
- Set as FOREIGN KEY and it should refer isbn in BOOKS table

```
CREATE DATABASE library;
USE library;
```

```
CREATE TABLE Branch (
    Branch_no INT PRIMARY KEY,
    Manager_Id INT,
    Branch_address VARCHAR(200),
    Contact_no VARCHAR(15)
);
```

```
DESC Branch;
```

Field	Type	Null	Key	Default	Extra
Branch_no	int	NO	PRI	NULL	
Manager_Id	int	YES		NULL	
Branch_address	varchar(200)	YES		NULL	
Contact_no	varchar(15)	YES		NULL	

```
CREATE TABLE Employee (
    Emp_Id INT PRIMARY KEY,
    Emp_name VARCHAR(100),
    Position VARCHAR(50),
    Salary DECIMAL(10,2),
    Branch_no INT,
    FOREIGN KEY (Branch_no)
    REFERENCES Branch(Branch_no)
);
DESC Employee;
```

Field	Type	Null	Key	Default	Extra
Emp_Id	int	NO	PRI	NULL	
Emp_name	varchar(100)	YES		NULL	
Position	varchar(50)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
Branch_no	int	YES	MUL	NULL	

```

CREATE TABLE Books (
    ISBN VARCHAR(20) PRIMARY KEY,
    Book_title VARCHAR(200),
    Category VARCHAR(50),
    Rental_Price DECIMAL(10,2),
    Status VARCHAR(5),
    Author VARCHAR(100),
    Publisher VARCHAR(100)
);
DESC Books;

```

Field	Type	Null	Key	Default	Extra
ISBN	varchar(20)	NO	PRI	NULL	
Book_title	varchar(200)	YES		NULL	
Category	varchar(50)	YES		NULL	
Rental_Price	decimal(10,2)	YES		NULL	
Status	varchar(5)	YES		NULL	
Author	varchar(100)	YES		NULL	
Publisher	varchar(100)	YES		NULL	

```

CREATE TABLE Customer (
    Customer_Id INT PRIMARY KEY,
    Customer_name VARCHAR(100),
    Customer_address VARCHAR(200),
    Reg_date DATE
);
DESC Customer;

```

Field	Type	Null	Key	Default	Extra
Customer_Id	int	NO	PRI	NULL	
Customer_name	varchar(100)	YES		NULL	
Customer_address	varchar(200)	YES		NULL	
Reg_date	date	YES		NULL	

```

CREATE TABLE IssueStatus (
    Issue_Id INT PRIMARY KEY,
    Issued_cust INT,
    Issued_book_name VARCHAR(200),
    Issue_date DATE,
    Isbn_book VARCHAR(20),

    FOREIGN KEY (Issued_cust)
    REFERENCES Customer(Customer_Id),

    FOREIGN KEY (Isbn_book)
    REFERENCES Books(ISBN)
);
DESC IssueStatus;

```

Field	Type	Null	Key	Default	Extra
Issue_Id	int	NO	PRI	NULL	
Issued_cust	Issue_Id	YES	MUL	NULL	
Issued_book_name	varchar(200)	YES		NULL	
Issue_date	date	YES		NULL	
Isbn_book	varchar(20)	YES	MUL	NULL	

- **CREATE TABLE** ReturnStatus (
 Return_Id **INT PRIMARY KEY**,
 Return_cust **VARCHAR(100)**,
 Return_book_name **VARCHAR(200)**,
 Return_date **DATE**,
 Isbn_book2 **VARCHAR(20)**,

 FOREIGN KEY (Isbn_book2)
 REFERENCES Books(ISBN)
);
- **DESC** ReturnStatus;

Result Grid | Filter Rows: | Export: | Wrap Cell Content

Field	Type	Null	Key	Default	Extra
Return_Id	int	NO	PRI	NULL	
Return_cust	varchar(100)	YES		NULL	
Return_book_name	varchar(200)	YES		NULL	
Return_date	date	YES		NULL	
Isbn_book2	varchar(20)	YES	MUL	NULL	

```

70
71 • INSERT INTO Branch VALUES
72   (1,101,'Kochi','9876543210'),
73   (2,102,'Thrissur','9876543211'),
74   (3,103,'Calicut','9876543212');
75 • SELECT * FROM Branch;

```

Result Grid | Filter Rows: | Edit: |

	Branch_no	Manager_Id	Branch_address	Contact_no
▶	1	101	Kochi	9876543210
	2	102	Thrissur	9876543211
	3	103	Calicut	9876543212
★	NULL	NULL	NULL	NULL

INSERT INTO Employee VALUES

```
(101,'Rahul','Manager',70000,1),  
(102,'Anjali','Manager',68000,2),  
(103,'Arun','Manager',72000,3),  
(104,'Nikhil','Assistant',45000,1),  
(105,'Meera','Librarian',55000,1),  
(106,'Akhil','Assistant',40000,2),  
(107,'Sneha','Librarian',52000,2),  
(108,'Vivek','Assistant',48000,3),  
(109,'Neethu','Librarian',60000,3);  
SELECT * FROM Employee;
```

Emp_Id	Emp_name	Position	Salary	Branch_no
101	Rahul	Manager	70000.00	1
102	Anjali	Manager	68000.00	2
103	Arun	Manager	72000.00	3
104	Nikhil	Assistant	45000.00	1
105	Meera	Librarian	55000.00	1
106	Akhil	Assistant	40000.00	2
107	Sneha	Librarian	52000.00	2
108	Vivek	Assistant	48000.00	3
109	Neethu	Librarian	60000.00	3

INSERT INTO Books VALUES

```
('ISBN001','History of India','History',30,'yes','R.S Sharma','NCERT'),  
( 'ISBN002','Database Systems','Education',40,'yes','Korth','McGraw Hill'),  
( 'ISBN003','Python Basics','Education',35,'no','John Smith','Tech Press'),  
( 'ISBN004','World History','History',28,'yes','Peter John','Oxford'),  
( 'ISBN005','Science Today','Science',20,'yes','David Lee','Pearson'),  
( 'ISBN006','Physics Concepts','Science',22,'no','Albert Ray','Pearson'),  
( 'ISBN007','Ancient History','History',32,'yes','R.K Gupta','NCERT');  
SELECT * FROM Books;
```

ISBN	Book_title	Category	Rental_Price	Status	Author	Publisher
ISBN001	History of India	History	30.00	yes	R.S Sharma	NCERT
ISBN002	Database Systems	Education	40.00	yes	Korth	McGraw Hill
ISBN003	Python Basics	Education	35.00	no	John Smith	Tech Press
ISBN004	World History	History	28.00	yes	Peter John	Oxford
ISBN005	Science Today	Science	20.00	yes	David Lee	Pearson
ISBN006	Physics Concepts	Science	22.00	no	Albert Ray	Pearson
ISBN007	Ancient History	History	32.00	yes	R.K Gupta	NCERT

99 • INSERT INTO Customer VALUES

```

100      (1,'Asha','Kochi','2021-05-10'),
101      (2,'Ravi','Thrissur','2023-01-15'),
102      (3,'Maya','Calicut','2020-08-20'),
103      (4,'Arjun','Kochi','2022-09-01'),
104      (5,'Diya','Kannur','2021-12-15');

```

```
105 • SELECT * FROM Customer;
```

Result Grid				
		Filter Rows:	Edit:	
Customer_Id	Customer_name	Customer_address	Reg_date	
1	Asha	Kochi	2021-05-10	
2	Ravi	Thrissur	2023-01-15	
3	Maya	Calicut	2020-08-20	
4	Arjun	Kochi	2022-09-01	
5	Diya	Kannur	2021-12-15	

INSERT INTO IssueStatus VALUES

```

(1,1,'History of India','2023-06-10','ISBN001'),
(2,2,'Database Systems','2023-05-20','ISBN002'),
(3,3,'World History','2023-06-18','ISBN004'),
(4,4,'Python Basics','2024-01-05','ISBN003');
SELECT * FROM IssueStatus;

```

Issue_Id	Issued_cust	Issued_book_name	Issue_date	Isbn_book
1	1	History of India	2023-06-10	ISBN001
2	2	Database Systems	2023-05-20	ISBN002
3	3	World History	2023-06-18	ISBN004
4	4	Python Basics	2024-01-05	ISBN003

```

4 • INSERT INTO ReturnStatus VALUES
5   (1,'Asha','History of India','2023-07-01','ISBN001'),
5   (2,'Ravi','Database Systems','2023-06-01','ISBN002');
7 • SELECT * FROM ReturnStatus;

```

Return_Id	Return_cust	Return_book_name	Return_date	Isbn_book2
1	Asha	History of India	2023-07-01	ISBN001
2	Ravi	Database Systems	2023-06-01	ISBN002

Display all the tables and Write the queries for the following :

1. Retrieve the book title, category, and rental price of all available books.

```

119 • SELECT Book_title, Category, Rental_Price
120 FROM Books
121 WHERE Status='yes';

```

Book_title	Category	Rental_Price
History of India	History	30.00
Database Systems	Education	40.00
World History	History	28.00
Science Today	Science	20.00
Ancient History	History	32.00

2. List the employee names and their respective salaries in descending order of salary.

```

23 • SELECT Emp_name, Salary
24 FROM Employee
25 ORDER BY Salary DESC;

```

Result Grid   Filter Rows:	
Emp_name	Salary
Arun	72000.00
Rahul	70000.00
Anjali	68000.00
Neethu	60000.00
Meera	55000.00
Sneha	52000.00
Vivek	48000.00
Nikhil	45000.00
Akhil	40000.00

3. Retrieve the book titles and the corresponding customers who have issued those books.

```

27 • SELECT b.Book_title,
28         c.Customer_name
29 FROM IssueStatus i
30 JOIN Books b
31 ON i.Isbn_book=b.ISBN
32 JOIN Customer c
33 ON i.Issued_cust=c.Customer_Id;

```

Result Grid   Filter Rows:	
Book_title	Customer_name
History of India	Asha
Database Systems	Ravi
World History	Maya
Python Basics	Arjun

4. Display the total count of books in each category.

```

.35 • SELECT Category,
.36         COUNT(*) AS Total_Books
.37     FROM Books
.38     GROUP BY Category;

```

Result Grid   Filter Rows: Ex	
Category	Total_Books
History	3
Education	2
Science	2

5. Retrieve the employee names and their positions for the employees whose salaries are above Rs.50,000.

```

.40 • SELECT Emp_name,
.41         Position
.42     FROM Employee
.43     WHERE Salary > 50000;

```

Result Grid   Filter Rows:	
Emp_name	Position
Rahul	Manager
Anjali	Manager
Arun	Manager
Meera	Librarian
Sneha	Librarian
Neethu	Librarian

6. List the customer names who registered before 2022-01-01 and have not issued any books yet.

```

5 • SELECT Customer_name
6     FROM Customer
7     WHERE Reg_date < '2022-01-01'
8     AND Customer_Id NOT IN
9     (
10    SELECT Issued_cust
11    FROM IssueStatus
12    );

```

ult Grid		Filter Rows:	
Customer_name			
Diya			

7. Display the branch numbers and the total count of employees in each branch.

```

4 • SELECT Branch_no,
5         COUNT(*) AS Employee_Count
6     FROM Employee
7     GROUP BY Branch_no;

```

ult Grid		Filter Rows:		Ex
Branch_no	Employee_Count			
1	3			
2	3			
3	3			

8. Display the names of customers who have issued books in the month of June 2023.

```

9 • SELECT DISTINCT c.Customer_name
10 FROM Customer c
11 JOIN IssueStatus i
12 ON c.Customer_Id=i.Issued_cust
13 WHERE MONTH(i.Issue_date)=6
14 AND YEAR(i.Issue_date)=2023;

```

Customer_name
Asha
Maya

9. Retrieve book_title from a book table containing history.

```

56 • SELECT Book_title
57 FROM Books
58 WHERE Book_title LIKE '%History%';

```

Book_title
History of India
World History
Ancient History

10. Retrieve the branch numbers along with the count of employees for branches having more than 5 employees

```

9 • SELECT Branch_no,
10 COUNT(*) AS Employee_Count
11 FROM Employee
12 GROUP BY Branch_no
13 HAVING COUNT(*) > 5;

```

Branch_no	Employee_Count
-----------	----------------

11. Retrieve the names of employees who manage branches and their respective branch addresses.

```
6 • SELECT e.Emp_name,  
7         b.Branch_address  
8 FROM Employee e  
9 JOIN Branch b  
0 ON e.Emp_Id=b.Manager_Id;
```

Result Grid	
Emp_name	Branch_address
Rahul	Kochi
Anjali	Thrissur
Arun	Calicut

12. Display the names of customers who have issued books with a rental price higher than Rs. 25.

```
2 • SELECT DISTINCT c.Customer_name  
3 FROM Customer c  
4 JOIN IssueStatus i  
5 ON c.Customer_Id=i.Issued_cust  
6 JOIN Books b  
7 ON i.Isbn_book=b.ISBN  
8 WHERE b.Rental_Price > 25;
```

Result Grid	
Customer_name	
Asha	
Ravi	
Arjun	
Maya	